

ISHAN BIJUKUCHHAY

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EDUCATION

Queen's University - Bachelor of Applied Science, Computer Engineering September 2025 - April 2030

PROJECT EXPERIENCE

Bare-Metal ARM Firmware (STM32F401RE, Cortex-M4) August 2026 - Present
Personal Project

- Wrote a bare-metal driver stack in C with no HAL or generated code: custom linker script, vector table, and C runtime startup (`.data` flash-to-RAM copy, `.bss` zeroing) derived from the reference manual.
- Implemented register-level GPIO and a UART transmit driver, deriving the fractional baud divisor from the clock tree and configuring alternate-function pin routing.
- Built interrupt-driven UART receive backed by a single-producer/single-consumer ring buffer requiring no critical sections (NVIC configuration, power-of-two index masking, drop-on-full overrun policy).
- Verified at three levels: GDB register reads, disassembly of the flashed binary, and logic-analyzer protocol decode; overflow behaviour matched predicted byte counts within 1%.

HTTP Caching Reverse Proxy July - August 2026
Personal Project

- Built an HTTP caching reverse proxy in C on a Raspberry Pi 5 (hash table with separate chaining, $O(1)$ lookup; doubly-linked LRU list, $O(1)$ eviction).
- Benchmarked on hardware with `wrk`: $\sim 46k$ req/s on cache hits vs $\sim 1.6k$ on origin fetches ($\sim 29\times$ throughput).
- Profiled concurrency tradeoffs and eviction policies (LRU held 75% hit rate vs LFU's 9.5% under drifting load).

Conveyor-Scoring Robot September 2025 - Present
Queen's VEX Robotics Club

- Wrote C++ for sensor-driven autonomous behaviour on an intake-and-conveyor scoring robot, working alongside the mechanical subteam.
- Integrated sensors and debugged signal paths between control electronics and actuators.

EcoTrack - Carbon Footprint Tracker 2026
Team Lead, APSC 103

- Led a team of 5 to ship a full-stack web app (React, Flask) converting everyday choices into CO₂e estimates; managed deadlines and client communication through to a working prototype.

WORK EXPERIENCE

Lifeguard/Instructor June 2025 - September 2025
City of Kitchener - Aquatics Division

- Planned, delivered, and adapted over 20 swim programs for 100+ participants across age groups.

EXTRACURRICULAR EXPERIENCE

General Manager September 2022 - June 2025
Huron Heights S.S. Robotics Team

- Led and coordinated 15+ project teams building battle bots, managing schedules to ensure on-time delivery.
- Assigned members to subteams by skill and collaboration dynamics; ran competition rules and logistics briefings.

SKILLS AND INTERESTS

Languages	C, C++, Rust, Python
Embedded	Bare-metal ARM Cortex-M4, UART, interrupt handling & NVIC, linker scripts
Systems	Data Structures & Algorithms, Multithreading, Sockets, Linux
Tools	Git, GDB, Make, GCC/arm-none-eabi, OpenOCD, logic analyzer (sigrok)
Hardware	PCB Design (Proteus), Fusion 360